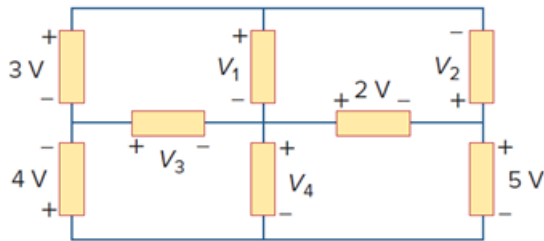


Problem

Given the circuit in Fig. 2.78, use KVL to find the branch voltages V_1 to V_4 .

Figure 2.78:



Step-by-step solution

Step 1 of 5

Refer to Figure 2.78 in the textbook.

Redraw the circuit as shown in figure 1.

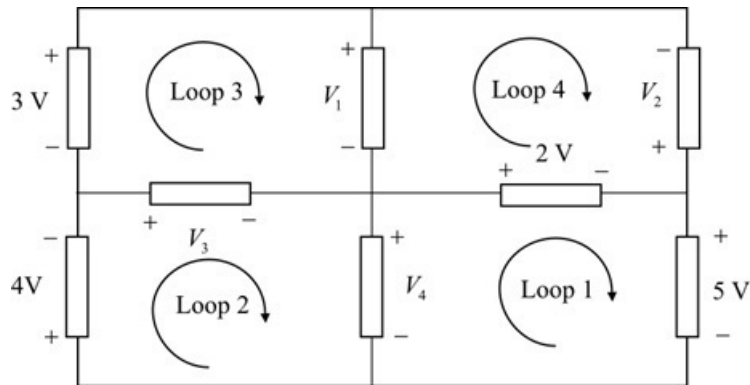


Figure 1

Step 2 of 5

Apply Kirchhoff's voltage law around the loop 1.

$$-V_4 + 2 + 5 = 0$$

$$-V_4 + 7$$

$$V_4 = 7 \text{ V}$$

Therefore, the voltage V_4 is 7 V.

Step 3 of 5

Apply Kirchhoff's voltage law around the loop 2.

$$4 + V_3 + V_4 = 0$$

Substitute 7 for V_4 .

$$4 + V_3 + 7 = 0$$

$$V_3 + 11 = 0$$

$$V_3 = -11 \text{ V}$$

Therefore, the voltage V_3 is $\boxed{-11 \text{ V}}$.

Step 4 of 5

Apply Kirchhoff's voltage law around the loop 3.

$$-3 + V_1 - V_3 = 0$$

Substitute -11 for V_3 .

$$-3 + V_1 - (-11) = 0$$

$$-3 + V_1 + 11 = 0$$

$$V_1 + 8 = 0$$

$$V_1 = -8 \text{ V}$$

Therefore, the voltage V_1 is $\boxed{-8 \text{ V}}$.

Step 5 of 5

Apply Kirchhoff's voltage law around the loop 4.

$$-V_1 - V_2 - 2 = 0$$

Substitute -8 for V_1 .

$$-(-8) - V_2 - 2 = 0$$

$$8 - V_2 - 2 = 0$$

$$-V_2 + 6 = 0$$

$$V_2 = 6 \text{ V}$$

Thus the voltage V_2 is $\boxed{6 \text{ V}}$.